ECA18 – OPTOELECTRONICS AND OPTICAL COMMUNICATION

CO2 PROBLEM SOLVING ASSIGNMENT

THE PROBLEM SOLVING ENGINEERING SITUATION

A communication system designer must develop an optical transmitter for an 8-channel DWDM fiber-optic link operating around the 1550 nm window with 100 GHz channel spacing. Each channel carries 25 Gb/s NRZ data over six amplified 80 km spans of single-mode fiber. The transmitter must provide adequate output power, high modulation speed, narrow linewidth, stable wavelength under ambient-temperature variation, and compliance with optical-interface standards and eye-safety limits. Candidate sources include an LED, Fabry–Perot (FP), DBR, DFB, VCSEL and quantum-dot (QD) lasers. The designer must then evaluate the nonlinear impairments (SPM, XPM, FWM, SBS and SRS), decide whether soliton transmission or digital backpropagation is appropriate, and select a discrete or photonic-integrated (InP or Si-photonics) transmitter implementation.

The proposed problem-solving study must satisfy the following information and constraints:

The objective is to design an optical transmitter for an 8-channel Dense Wavelength Division Multiplexing (DWDM) system operating near 1550 nm with 100 GHz channel spacing. Each channel carries 25 Gb/s NRZ data through six amplified spans of 80 km standard single-mode fiber, giving a total transmission distance of 480 km.

The transmitter must provide sufficient optical power, high modulation bandwidth, narrow linewidth, stable wavelength, low nonlinear distortion, and compliance with relevant optical-interface and laser-safety requirements.

## Stage 1 – Interpret the communication requirement

### 1.1 Main requirements

| Parameter | Specification |
|---|---|

| | |
|---|---|
| Number of channels | 8 |
| Operating wavelength | Approximately 1550 nm |
| Channel spacing | 100 GHz ≈ 0.8 nm |
| Data rate per channel | 25 Gb/s |
| Total raw data rate | 200 Gb/s |
| Total fiber distance | 480 km |
| Fiber attenuation | 0.2 dB/km |
| Launch power per channel | 2 mW |
| Total launch power | 16 mW |
| Wavelength tolerance | ±0.1 nm |
| Temperature variation | 40°C |

## 1.2 Required transmitter characteristics

- High modulation speed: supports the required 25 Gb/s NRZ data rate.
- Narrow linewidth: reduces phase noise and helps limit stimulated Brillouin scattering (SBS) sensitivity.
- High side-mode suppression: reduces unwanted spectral components and adjacent-channel interference.
- Stable wavelength: prevents the optical carrier from drifting away from its assigned DWDM channel.
- Controlled chirp: reduces the interaction between modulation and fiber dispersion.
- Controlled output power: balances receiver performance and nonlinear distortion.

- Reliable integration: enables compact, efficient, and scalable multichannel operation.

### 1.3 Standards and safety

- ITU-T G.957: provides optical-interface specifications for applicable SDH systems; it is not a universal specification for every DWDM transmitter.
- IEEE 802.3bs/bm: provides specifications for particular high-speed Ethernet PHYs. Compliance must be checked against the exact interface and wavelength configuration; these standards do not automatically define the proposed 8-channel DWDM system.
- ANSI Z136: provides laser-safety requirements. The applicable exposure limit depends on wavelength, accessible emission, exposure duration, and optical configuration.

Problem requirement statement: Design an eight-channel, 25 Gb/s-per-channel DWDM transmitter near 1550 nm with 100 GHz channel spacing, controlled output power and chirp, narrow spectral linewidth, wavelength stability of ±0.1 nm, and suitable nonlinear compensation and optical safety measures for a 480 km amplified link.

# Stage 2 – Investigate and select the optical source

### 2.1 Emission wavelength

Given Eg = 0.80 eV.

$$\lambda \approx 1.24 / Eg$$

$$= 1.24 / 0.80$$

$$= 1.55\ \mu m$$

$$= 1550\ nm$$

This wavelength is suitable for the low-attenuation region of standard single-mode optical fiber.

### 2.2 Internal quantum efficiency

Given radiative lifetime $\tau r$ = 10 ns and non-radiative lifetime $\tau nr$ = 40 ns.

$$\eta i = \tau nr / (\tau r + \tau nr)$$

$$= 40 / (10 + 40)$$

$$= 0.80 = 80\%$$

Under the stated lifetime model, 80% of recombination events are radiative.

### 2.3 Comparison of optical sources

| Source | Characteristics | Suitability |
|---|---|---|
| LED | Broad spectrum; lower power and limited modulation speed | Unsuitable for long-haul DWDM |
| Fabry–Perot laser | Multiple longitudinal modes; useful power and speed | Less suitable due to broader spectrum |
| DBR laser | Narrow, wavelength-selective output; high speed | Good alternative |
| DFB laser | Narrow linewidth, strong single-mode operation, useful power | Excellent for DWDM |
| VCSEL | Compact and efficient; design-dependent wavelength and coupling | Better for suitable short-reach links |
| Quantum-dot laser | Potential low threshold and temperature-related advantages | Promising, depending on implementation |

Laser oscillation begins when optical gain equals or exceeds total resonator losses. Below threshold, spontaneous emission dominates; above threshold, stimulated emission produces coherent output. Linewidth describes spectral width, while side-mode suppression ratio (SMSR) compares the main mode with unwanted modes.

Design Decision 1: I recommend a 1550 nm DFB laser for each DWDM channel because it provides narrow spectral output, high modulation capability, stable single-mode operation, and suitable fiber coupling. A DBR laser is a credible alternative if its tuning range and wavelength control better suit the implementation.

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## Stage 3 – Modulation, Chirp and Spectral Stability

### 3.1 Output power

Given threshold current Ith = 12 mA, slope efficiency ηs = 0.3 mW/mA, and bias current I = 40 mA.

$$P = \eta s(I - Ith)$$

$$= 0.3(40 - 12)$$

$$= 8.4\ mW$$

$$P(dBm) = 10\ log10(8.4) = 9.24\ dBm$$

This is the calculated laser output at the stated bias current. Actual fiber launch power depends on modulator, coupling and other insertion losses.

## 3.2 Modulation bandwidth

$$B \approx 0.7Rb = 0.7 \times 25\ Gb/s = 17.5\ GHz$$

$$Bandwidth\ margin = 20 - 17.5 = 2.5\ GHz$$

$$Percentage\ margin = (2.5 / 17.5) \times 100 \approx 14.3\%$$

The candidate 20 GHz bandwidth meets the estimated requirement. The modulator, driver, receiver and complete link bandwidth still need verification.

## 3.3 Carrier and photon rate equations

$$dN/dt = I/(qV) - N/\tau n - G(N,S)S$$

$$dS/dt = \Gamma G(N,S)S + \beta spN/\tau r - S/\tau p$$

N is carrier density; S is photon density; q is electron charge; V is active-region volume; G(N,S) is optical gain; Γ is confinement factor; βsp is the spontaneous-emission coupling factor; τn is carrier lifetime; and τp is photon lifetime. Relaxation oscillations arise from coupled carrier–photon dynamics. Small-signal modulation is analyzed around the bias point, while large-signal digital switching can produce nonlinear transients, overshoot and pattern-dependent effects. The exact relaxation-oscillation frequency cannot be calculated from the supplied 20 GHz bandwidth alone.

## 3.4 Direct versus external modulation

| Feature | Direct modulation | External modulation |
|---|---|---|
| Principle | Injection current directly modulates the laser | Separate modulator controls light from a CW laser |
| Cost/complexity | Lower | Higher |
| Chirp | Can be significant | Can be lower with a suitable modulator |
| Dispersion interaction | May increase pulse distortion | Better suited to long-distance transmission |
| Suitability | Simpler or shorter links | Preferred for the proposed multispan link |

A Mach–Zehnder modulator (MZM) is recommended for external modulation to reduce chirp-related penalties over 480 km.

### 3.5 Wavelength drift and temperature control

$\Delta\lambda = 0.1\ \text{nm/°C} \times 40\text{°C} = 4\ \text{nm}$

The uncontrolled drift is five times the approximate 0.8 nm channel spacing and 40 times the permitted 0.1 nm deviation.

$\Delta T_{max} = 0.1\ \text{nm} / (0.1\ \text{nm/°C}) = \pm 1\text{°C}$

This is an approximate temperature-stability requirement based on the supplied coefficient. A thermoelectric cooler (TEC) is recommended for stable wavelength control. Injection-current tuning can be faster but also affects optical power and chirp.

Design Decision 2: I recommend external MZM modulation and TEC-based wavelength stabilization, with the laser operated at a controlled bias current.

## Stage 4 – Nonlinear Impairments and Control

### 4.1 Effective fiber length

The attenuation must be converted from dB/km to nepers/km.

$\alpha = 0.2 / 4.343 = 0.04605\ \text{km}^{-1}$

$L_{eff} = (1 - e^{\wedge}(-\alpha L)) / \alpha$

$L_{eff} = (1 - e^{\wedge}(-0.04605 \times 80)) / 0.04605 \approx 21.3\ \text{km}$

### 4.2 Self-phase modulation

$\varphi_{NL} = \gamma P L_{eff}$

$= 1.3 \times 0.002 \times 21.3$

$\approx 0.0554\ \text{rad/span}$

For six spans: $\varphi_{NL,total} \approx 6 \times 0.0554 = 0.332\ \text{rad}$

This estimate assumes the same effective power conditions in each span and neglects detailed amplifier and dispersion interactions. SPM can cause spectral broadening and interact with chromatic dispersion.

### 4.3 XPM and FWM

Cross-phase modulation (XPM) occurs when the intensity of one channel changes another channel's phase through the Kerr effect. Four-wave mixing (FWM) generates new frequency components

through interaction among optical channels. These effects depend on channel powers, spacing, fiber dispersion, phase matching and polarization. Generated components can cause crosstalk when they overlap existing DWDM channels.

## 4.4 SBS and SRS thresholds

Given Aeff = $80 \times 10^{-12}$ $m^2$ and Leff = 21.3 km = 21,300 m.

$$P_{th,SBS} \approx 21A_{eff} / (gBL_{eff})$$

$$= 21(80 \times 10^{-12}) / [(5 \times 10^{-11})(21300)] \approx 1.58 \text{ mW}$$

$$P_{th,SRS} \approx 16A_{eff} / (gRL_{eff})$$

$$= 16(80 \times 10^{-12}) / [(6 \times 10^{-14})(21300)] \approx 1.00 \text{ W}$$

| Effect | Estimated threshold | Comparison | Assessment |
|---|---|---|---|
| SBS | 1.58 mW | 2 mW per channel | Potentially significant |
| SRS | Approximately 1 W | 16 mW total | Below estimated threshold |

These are approximate estimates from the supplied equations. Actual thresholds depend on linewidth, polarization, fiber properties and power distribution. SBS arises from interaction with acoustic phonons and tends to backscatter light; SRS arises from optical phonons and can transfer power to shifted frequencies. Increasing laser linewidth can raise the SBS threshold, subject to modulation and receiver requirements.

## 4.5 Eye safety

$$P_{total} = 8 \times 2 \text{ mW} = 16 \text{ mW}$$

$$P_{total(dBm)} = 10 \log_{10}(16) \approx 12.04 \text{ dBm}$$

For a preliminary illustration only, assume a hypothetical accessible-emission screening value of 10 mW. The 16 mW total exceeds this assumed value. This is not the actual ANSI Z136 permissible exposure limit or a laser classification. A formal assessment must consider accessible emission, beam geometry, wavelength, exposure duration and the complete optical system. Use enclosed connections, appropriate interlocks, warning labels and safe maintenance procedures.

## 4.6 Nonlinearity mitigation and decision

Optimize channel launch power and amplifier gain; use span-by-span power equalization while maintaining adequate receiver sensitivity.

Use dispersion-aware DSP and, where beneficial, nonlinear compensation to reduce combined dispersion and Kerr effects.

## Stage 5 – Soliton Transmission, Digital Backpropagation and Integration

### 5.1 Dispersion length

$$LD = T0^2 / |\beta 2| = (15\ ps)^2 / (20\ ps^2/km) = 11.25\ km$$

Soliton propagation requires a balance between anomalous chromatic dispersion and Kerr nonlinearity.

### 5.2 Fundamental-soliton peak power

$$\text{For } N = 1,\ LD = LNL \text{ and } P0 = 1/(\gamma LD)$$

$$P0 = 1/[1.3 \times 11.25] \approx 0.0684\ W = 68.4\ mW$$

### 5.3 Nonlinear length and soliton order

$$LNL = 1/(\gamma P0) = 1/[1.3 \times 0.002] \approx 384.6\ km$$

$$N = \sqrt{(LD/LNL)} = \sqrt{(11.25/384.6)} \approx 0.171$$

The specified 2 mW launch power does not satisfy the fundamental-soliton condition under this simplified model. Soliton transmission is therefore not recommended. Raising launch power would require reassessment of SBS, other nonlinearities, amplifier behaviour and safety.

### 5.4 Digital backpropagation

Digital backpropagation (DBP) numerically approximates reverse fiber propagation to compensate for chromatic dispersion and Kerr nonlinearities. It can improve signal quality when the fiber model is accurate, but computational complexity grows with model accuracy. Noise added by optical amplifiers cannot simply be reversed, and single-channel DBP does not fully compensate for interchannel XPM and FWM. DBP may be evaluated at the receiver if its measured benefit justifies the complexity.

### 5.5 Standards verification

| Standard | Relevance | Verification |
|---|---|---|
| ITU-T G.957 | Optical interfaces for applicable SDH systems | Check the exact interface specification |
| IEEE 802.3bm | Certain 40G/100G Ethernet PHYs | Check relevant PHY requirements if used |
| IEEE 802.3bs | Certain 200G/400G Ethernet PHYs | Check selected PHY and optical parameters |
| ANSI Z136 | Laser safety | Assess accessible emission and applicable exposure limits |

The 200 Gb/s aggregate raw data rate alone does not establish compliance with any Ethernet PHY. The proposed DWDM interface must be checked against the actual selected standard.

### 5.6 Photonic integration comparison

| Feature | Discrete | Monolithic InP | Si-photonics with hybrid laser |
|---|---|---|---|
| Footprint | Relatively large | Compact | Compact |
| Power | Can be higher | Design-dependent | Potentially efficient |
| Coupling | Many individual connections | Integrated optical path | Requires efficient laser-to-chip coupling |
| Reliability | More discrete interconnections | Fewer external optical connections | Depends on packaging and hybrid laser |
| Scalability | More assembly effort | Good active-device integration | Strong multichannel integration potential |
| Flexibility | High | Platform-dependent | High potential for active/passive integration |

Design Decision 4: I recommend silicon photonics with hybrid-integrated narrow-linewidth DFB lasers, external modulators, wavelength-selective multiplexing and TEC control. Use conventional DSP and evaluate DBP if its performance benefit warrants its complexity.

## Stage 6 – Verification and Critical Evaluation

| Requirement | Calculated/selected result | Assessment |
|---|---|---|
| Optical source | 1550 nm DFB laser per channel | Suitable, subject to component verification |
| Output power | 8.4 mW or 9.24 dBm at stated bias | Matches supplied laser model |
| Bandwidth | 17.5 GHz required; 20 GHz available | 2.5 GHz estimated margin |
| Wavelength stability | Uncontrolled drift 4 nm; target ±0.1 nm | Requires TEC and calibration |
| Nonlinear phase | 0.0554 rad/span; 0.332 rad over six spans (simplified) | Validate at link level |
| SBS | Estimated 1.58 mW threshold | Mitigation and measurement required |
| SRS | Estimated 1 W threshold | Below estimate at 16 mW total |
| Soliton | $N \approx 0.171$ | Not selected at specified launch power |
| DSP | Conventional DSP; optional DBP | Depends on performance and complexity |

| | | |
|---|---|---|
| Integration | Hybrid laser with silicon photonics | Recommended, subject to qualification |
| Standards and safety | Check applicable interface and laser safety requirements | Formal verification required |

The preliminary design meets the calculated wavelength and bandwidth targets. Wavelength stabilization, SBS management, end-to-end link performance and formal standards compliance remain key verification tasks.

## Proposed block diagram

**Eight independent 25 Gb/s NRZ data inputs**

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**Eight DFB laser sources near 1550 nm with TEC and bias control**

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**Eight external Mach–Zehnder modulators**

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**DWDM multiplexer – 100 GHz channel spacing**

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**Optical amplifier and six 80 km single-mode fiber spans**

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**DWDM demultiplexer and optical receivers**

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**DSP – dispersion compensation, optional DBP, BER and signal-quality monitoring**